Decision Support System for Sri Lankan Paddy Cultivation by Predicting disease stages and Aligning Cultivation & Harvest

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Project Proposal Report

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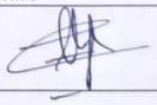
DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lankan paddy agriculture faces economic threats from fungal and bacterial diseases such as rice blast and brown spot which cause yield losses exceeding 30% in severely affected seasons. Existing AI-based disease detection tools available to Sri Lankan farmers only classify disease as present or absent and they provide no information about how serious the infection is, and no guidance on what the farmer should do about it. AI-based disease detection systems classify diseases as present or absent, providing no quantitative assessment of severity, no progression modeling over time. This absence of risk-informed decision support forces farmers to rely on expert guidance that is often unavailable in rural settings.

This component proposes integrated disease detection, severity grading, and management recommendation system localized for Sri Lankan paddy conditions. Using a locally collected paddy leaf image dataset annotated by certified Department of Agriculture Sri Lanka officers, deep learning models based on EfficientNet-B4 and CNN architectures are trained to detect two diseases rice blast, brown spot and grade their severity on a five-point scale from 0 (healthy) to 4 (critical). A rule-based recommendation engine built on DoA-SL crop management guidelines then translates each severity grade into a structured, plain-language management recommendation delivered in both English and Sinhala, specifying urgency level, recommended intervention, and the financial consequence of delayed action.

Expected outcomes include a disease classification model on Sri Lankan Field test data, severity on grading five levels, and end-to-end inference within five seconds, and recommendations rated appropriate by at least 80% of independent DoA-SL expert reviewers. This component directly contributes to reducing preventable yield losses and strengthening the financial resilience of Sri Lankan paddy farmers.

Keywords: *Deep Learning; Paddy Disease Detection; Severity Classification; Management Recommendations; EfficientNet-B4; Sri Lankan Agriculture; CNN; DoA-SL Guidelines.*

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I also want to thank my co-supervisor for the valuable domain knowledge shared on disease identification and system integration, which helped me keep the methodology connected to the realities of Sri Lankan paddy farming. My team members have been a constant source of support, and working alongside them on a shared system has made every individual component stronger.

I sincerely thank the Sri Lanka Institute of Information Technology for the academic environment, resources, and opportunities that made this research possible. I hope this work makes a meaningful contribution to Sri Lankan paddy agriculture.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
DSS	Decision Support System
DoA	Department of Agriculture Sri Lanka
EfficientNet	Efficient Neural Network (Scalable CNN Architecture)
IRRI	International Rice Research Institute
SES	Standard Evaluation System (IRRI)
FR	Functional Requirement
IoT	Internet of Things
LSTM	Long Short-Term Memory Network
ML	Machine Learning
MAP	Mean Average Precision
NPK	Nitrogen, Phosphorus, Potassium
RGB	Red Green Blue (colour image channels)
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SST	Software Systems and Technologies
REST API	Representational State Transfer Application Programming Interface

1.INTRODUCTION

1.1. Background and Context

Rice is not just a crop in Sri Lanka; it is a way of life. Approximately 1.8 million farming families across the country depend on paddy cultivation as their primary source of income, and rice accounts for roughly 45% of the average daily calorie intake of the Sri Lankan population [1]. The North Central Province districts of Anuradhapura and Polonnaruwa together contribute the largest share of the national paddy harvest, and their semi-arid climate creates growing conditions that are particularly prone to fungal and bacterial disease outbreaks during peak humidity months.

The two most destructive paddy diseases in Sri Lanka are rice blast, caused by the fungus Magnaporthe oryzae, and brown spot, caused by Bipolaris oryzae. According to the International Rice Research Institute, these diseases can reduce yields by between 10% and 50% depending on the crop growth stage at the time of infection and the speed of treatment response [2]. The Department of Agriculture Sri Lanka estimated that unmanaged rice blast outbreaks during the Maha season alone cost the country more than LKR 4.7 billion in annual crop losses [3]. Beyond the immediate harvest, a farmer who misses the critical treatment window in the first week of visible symptoms may also face residual soil contamination that damages the following cultivation cycle.

Current disease management in Sri Lanka relies almost entirely on visual inspection by trained agricultural extension officers. However, Pathirana and Wijesinghe (2020) documented that the ratio of agricultural officers to farmers in many rural districts exceeds 1:800, making timely expert-guided disease identification practically impossible for the vast majority of farming households [4]. Without professional guidance, farmers typically apply fungicides across the entire field as a precaution an approach that increases input costs, promotes pesticide resistance, and harms soil and water quality [5].

In recent years, deep learning and computer vision have opened a promising path toward automated disease detection. Perera, Karunaratne, and Senevirathne (2022) demonstrated that transfer learning models fine-tuned on locally collected Sri Lankan field imagery can achieve disease classification accuracy above 91% significantly outperforming models trained on generic international datasets [6]. However, even these more accurate systems share a fundamental limitation: they classify disease as present or absent and output a single disease label. They say nothing about how severe the infection is, which determines both the urgency of treatment and the scale of potential crop loss. For a smallholder farmer in Polonnaruwa whose entire household income depends on the pending harvest, a label that says 'blast detected' is not sufficient. What that farmer genuinely needs to know is: how serious is it right now, what should I do specifically, and what will it cost me if I wait another week?

This research component addresses exactly that gap. It moves beyond binary disease detection to deliver a three-stage output: disease identification, multi-class severity grading on a 0–4 scale, and a structured management recommendation that specifies urgency, intervention, and financial consequences all localized for Sri Lankan paddy varieties and conditions and accessible through a REST API integrated with the group's unified Decision Support System dashboard.

1.2 Literature Review

A growing body of research has explored the use of machine learning and deep learning for crop disease detection, and several studies are directly relevant to the approach taken in this component.

One of the earliest attempts to apply image processing to paddy disease identification in Sri Lanka came from Anthonys and Wickramarachchi (2009), who developed a recognition system using texture, colour, and shape features extracted from field-collected leaf images to distinguish rice blast, sheath blight, and brown spot [1]. The system used a membership function classifier and was built entirely on Sri Lankan paddy imagery, making it a foundational reference for locally grounded disease detection research. Its key limitation was that the approach predated deep learning entirely the feature engineering was manual and the classifier could not generalize to the level of accuracy that modern convolutional architectures achieve on the same task.

Pemasiri and Vidanagama (2022) built directly on this tradition by training CNN models on a dataset of 800 natural paddy leaf and stem images captured from Sri Lankan rice experimental fields [2]. Their study was notable for using entirely locally sourced imagery rather than supplementing with international datasets, and they reported classification accuracy across multiple disease categories that outperformed models trained on mixed datasets. The limitation of their work was the relatively small dataset size 800 images across multiple classes leaves certain severity presentations underrepresented, and the study did not attempt severity grading or any form of management recommendation output.

Jayasinghe and Ranasinghe (2021) applied a CNN to paddy leaf disease classification under Sri Lankan field conditions, collecting images from farms in the North Western Province and reporting 89.3% overall accuracy across three disease categories [3]. Their work confirmed that CNN-based detection is technically feasible in local growing conditions and that locally collected imagery improves generalisation over generic datasets. However, the system output only binary healthy-or-diseased labels, and no severity assessment or structured recommendation was produced.

Researchers from the University of Peradeniya (2023) published a study in PLOS ONE on a low-cost IoT monitoring system for polytunnels in Peradeniya, Sri Lanka, deploying temperature, humidity, and light sensors in a real agricultural environment and automating climate control responses based on sensor-driven thresholds [7]. Their work confirmed that IoT sensor deployment in Sri Lankan agricultural conditions is practically feasible and cost-

effective at the farm level, supporting the technical viability of the environmental integration approach adopted in this component.

Ekanayake, Weliwatta, and Jayasinghe (2021) from Wayamba University of Sri Lanka developed crop-weather models for paddy yield prediction covering seven geographical regions contributing roughly two-thirds of national paddy production, using nine weather indices including rainfall, relative humidity, temperature, wind speed, and sunshine hours [8]. Published in the Journal of Mathematics, their study established strong empirical relationships between environmental variables and yield outcomes in Sri Lankan paddy — relationships that underpin the rationale for connecting disease severity grades to expected harvest loss estimates in this component's recommendation outputs.

The e-Krushi portal maintained by the Department of Agriculture Sri Lanka provides web-based agricultural advisory services including disease identification guides, treatment schedules, and fertilizer recommendations [11]. While e-Krushi is a valuable government-backed resource and is widely known among extension officers, it is a static information system it does not offer real-time image-based detection, automated severity grading, or personalized management recommendations based on the specific disease and severity present in a farmer's field.

After reviewing both academic studies and commercial products, three limitations emerge that collectively justify the novelty of this component:

1. No existing system provides a fine-grained, multi-class severity grade for Sri Lankan paddy diseases. All reviewed systems produce binary outputs diseased or healthy or at best a coarse two-level classification. This makes it impossible for a farmer to understand how urgent their situation is or to make an informed decision about treatment timing.

2. No system translates a detected disease into a consequence-aware, urgency-graded management recommendation that specifies what to do, how quickly to do it, and what happens financially if action is delayed. Farmers receive a disease label, not a decision.

3. No system has been specifically trained and validated on Sri Lankan paddy disease morphology using images collected from Sri Lankan fields under local growing conditions and annotated by Department of Agriculture Sri Lanka certified officers.



Figure 01: Rice blast



Figure 01: Brown Spot

Table 2: literature Review

Author(s)	**Institution / Source**	**Contribution**	**Key Limitation**
Anthonys & Wickramarachchi (2009)	University of Moratuwa	First Sri Lankan image-based paddy disease identification system using local field data	Pre-deep learning; manual feature engineering; no severity grading
Pemasiri & Vidanagama (2022)	Sri Lankan rice experimental fields	Locally sourced Sri Lankan field imagery; multi-disease classification	Small dataset (800 images); no severity grading; no recommendation output
Jayasinghe & Ranasinghe (2021)	Sri Lankan farms, North Western Province / Ceylon Journal of Science	89.3% accuracy on locally collected Sri Lankan field images	Binary healthy/diseased only; no severity grade; no recommendation
University of Peradeniya researchers (2023)	University of Peradeniya	Low-cost IoT deployment validated in Sri Lankan agricultural environment	Limited to climate control; no connection to disease detection or severity
Ekanayake, Weliwatta & Jayasinghe (2021)	Wayamba University of Sri Lanka / Journal of Mathematics (Wiley)	Strong weather–yield relationships across 7 Sri Lankan paddy regions	No disease severity as input variable; yield prediction only
Department of Agriculture Sri Lanka (e-Krushi) (2024)	Government portal	Government-validated disease guides and treatment information	Static system; no real-time image detection; no personalized recommendation

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate an AI-based paddy disease severity estimation system that integrates deep learning leaf image analysis with real-time environmental sensor data, producing multi-class severity classifications, disease progression forecasts and actions management recommendations localized for Sri Lankan paddy cultivation conditions, accessible through a deployable REST API.

2.2 Specific Objectives

Dataset Collection and Labeling: Collect and label a Sri Lanka-specific paddy leaf image dataset covering the four most prevalent rice diseases in the local context rice blast and brown spot. Images will be categorized across five severity grades (0 = Healthy, 1 = Mild, 2 = Moderate, 3 = Severe, 4 = Critical) following IRRI SES protocols, targeting a minimum of 500 images per disease class, annotated by certified DoA-SL agricultural officers with dual verification.

Deep Learning Severity Classification: Train and evaluate CNN and EfficientNet-based deep learning models for multi-class disease severity classification. The models will be developed and validated on the Sri Lanka-localized dataset collected under SO1, with a target performance benchmark on the held-out Sri Lanka-specific test data, ensuring that classification reliability meets the threshold required for practical deployment in paddy field conditions.

Environmental Data Integration: Integrate real-time IoT sensor data specifically humidity, temperature, and leaf wetness readings with image-based disease severity outputs in order to model disease progression rates under current field conditions. This integration enables the system to move beyond static severity snapshots toward a dynamic, environmentally informed prediction of how a detected disease is likely to escalate over time.

Management Recommendation Engine: Generate actionable, risk-informed management recommendations based on the combined outputs of the severity classifier, the progression model. These recommendations will specify the urgency of required treatment, the recommended intervention type, and a clear articulation of the projected loss consequences if corrective action is delayed, giving farmers a decision-ready output rather than raw analytical data.

System Integration and API: Expose all disease detection, severity grading, and recommendation outputs through a FastAPI REST API that integrates with the group's shared DSS dashboard and stores all inference results, timestamps, and farm identifiers in PostgreSQL for trend analysis and farm-level history retrieval.

3. METHODOLOGY

3.1 Research Approach

This component follows the Design Science Research (DSR) methodology, which is well suited to research whose goal is to design, build, and evaluate a functioning technical artefact that solves a real-world problem (Hevner et al., 2004). The six-activity DSR model by Peffers et al. (2007) structures the work across problem identification (Section 1), objective definition (Section 2), artefact design and development (this section), demonstration, evaluation, and communication (Sections 4–5 and the final dissertation).

DSR was preferred over a purely experimental or survey-based approach because the aim of this research is not simply to study disease detection as a phenomenon, it is to build a working system, deploy it in real field conditions, and demonstrate measurable improvement over what currently exists for Sri Lankan paddy farmers.

3.2 Tools and Technologies

The entire system is developed using Python 3.11 as the primary programming language, chosen for its extensive ecosystem of deep learning, computer vision, and scientific computing libraries. TensorFlow 2.x with the Keras API provides the deep learning framework for model training, transfer learning, and inference, selected for its production-grade stability and strong support for EfficientNet architectures.

EfficientNet-B4 is selected as the primary model architecture. Its compound scaling approach simultaneously scaling network depth, width, and input resolution in a principled wa has consistently demonstrated superior generalization to field image variation compared to fixed-architecture alternatives such as ResNet and VGG. Transfer learning from ImageNet pre-trained weights is applied, with the classification head replaced and fine-tuned on the Sri Lankan dataset through a two-stage training process: the backbone is frozen for the first ten epochs to establish stable feature representations, then the full network is fine-tuned with a lower learning rate. A custom CNN baseline with four convolutional blocks, batch normalisation, and dropout is trained from scratch in parallel as a computational reference point.

OpenCV handles all image preprocessing operations including resizing, normalisation, and contour-based leaf region segmentation prior to model inference. During inference, five-crop Test-Time Augmentation (TTA) is applied to improve prediction stability when images are captured at angles or under uneven field lighting. The Adam optimizer with cosine annealing learning rate scheduling is used for training, with cross-entropy loss for the multi-class classification head and mean squared error for the continuous severity regression head. Data augmentation during training includes random rotation up to 30 degrees, horizontal flip, brightness and contrast jitter, and Gaussian noise to simulate the variability of field photography conditions.

The rule-based recommendation engine is implemented in Python using DoA-SL crop management guidelines as the knowledge base. It maps each combination of disease type and severity grade to a structured output record: an urgency level drawn from four tiers (Monitor, Act within 72 hours, Act within 24 hours, Critical), a recommended fungicide or cultural practice from the DoA-SL approved input list, and a consequence statement describing the financial risk of delayed action. This logic is implemented as a rule table rather than a trained model, which makes it fully transparent, auditable, and updatable as DoA-SL guidelines evolve.

FastAPI provides the REST API layer, offering high-performance asynchronous request handling that enables real-time inference responses. The API is containerized using Docker and deployed on AWS ECS with horizontal auto-scaling triggered at 70% CPU utilization. PostgreSQL stores all inference results, severity scores, recommendation outputs, and timestamps, with row-level security ensuring each farm can access only its own records. The farmer-facing interface is built in React.js, consistent with the frontend stack used across the group project. All data in transit is encrypted using TLS 1.3, and data at rest uses AES-256 encryption.

3.3 Data Collection and Ethical Considerations

Primary image data is collected in collaboration with local paddy farmers in the North Central Province. Written informed consent is obtained from all participating farmers before any data collection begins, and no personally identifying information is stored alongside images. Field visits are conducted with prior approval from the relevant Divisional Secretariat and Regional Agricultural Director. Severity labels are assigned by certified DoA-SL agricultural officers, not by the researcher alone, to ensure clinical accuracy. All data is stored in secured AWS S3 with access restricted to the research team and supervisor. Farm location data is recorded at district level only GPS coordinates are not stored to protect farmer privacy. Data collection procedures comply with the Sri Lanka Computer Crimes Act (2007) and SLIIT Faculty of Computing Research Committee ethical clearance guidelines.

3.4 Validation Strategy and Performance Metrics

Table 2: Validation Strategy

Metric	Target	Rationale
Severity Regression MAE	≤ 0.10	Mean absolute error on 0–1 scale; <0.10 corresponds to sub-grade precision following Lu et al. [12] benchmark.
Inference Latency	≤ 5 seconds	End-to-end time from image submission to recommendation display; targets real-time field usability on mobile.
Localization Accuracy Gain	≥ +8% F1 vs generic model	Quantifies the benefit of Sri Lanka-specific fine-tuning over a model trained only on international datasets.
Expert Agreement Rate	≥ 80%	Proportion of recommendations rated 'appropriate' by two independent agricultural experts during validation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall System Architecture

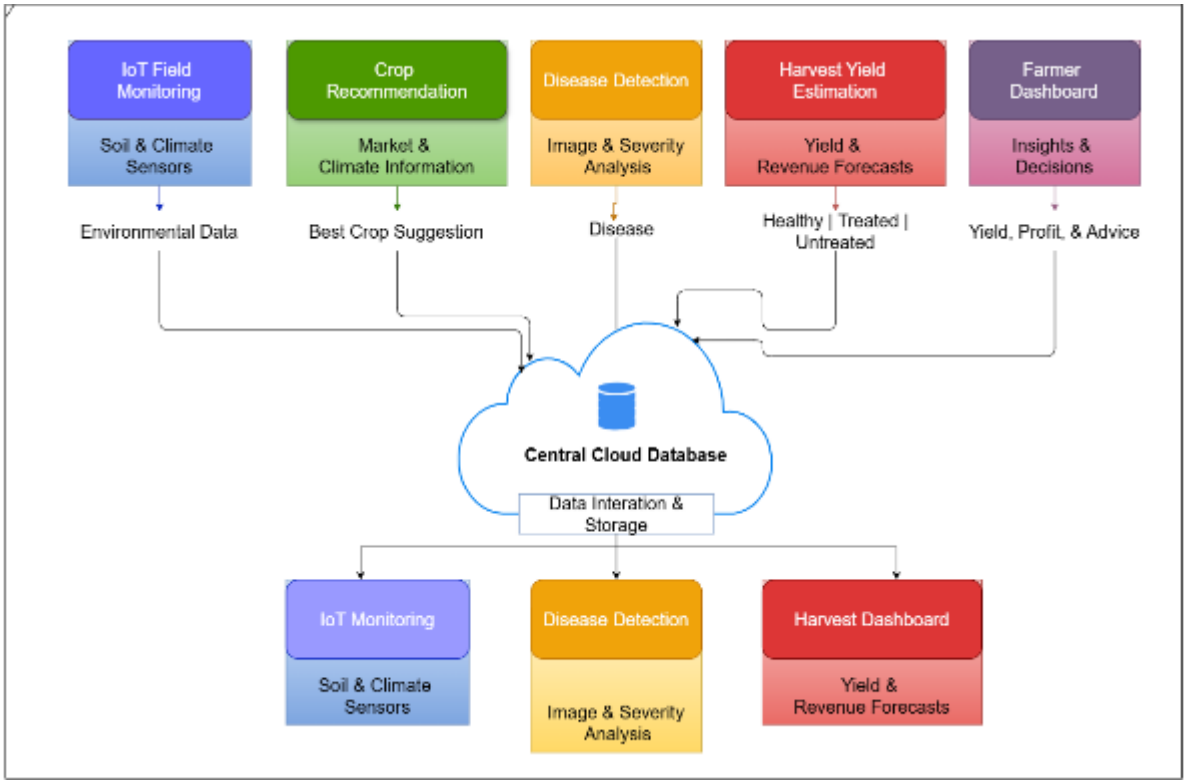


Figure 3: overall System Diagram

The overall system consists of four interconnected microservices deployed behind a unified API gateway on AWS:

- **IoT Field Monitoring** (IT22178718): Continuously collects real-time soil and environmental data pH, NPK levels, moisture, humidity, temperature, rainfall, leaf wetness, and water level from sensors deployed in paddy fields. Exposes a live sensor data API consumed by all downstream components.
- **Market-Aware Crop Recommendation** (IT21811104): Forecasts paddy market supply and demand trends and integrates IoT field data to recommend economically optimal crops before cultivation begins, fusing agronomic suitability with market profitability analysis.
- **Disease Detection** (IT22261632 this component): Accepts paddy leaf images from the farmer; classifies disease type; grades severity on a 0–4 scale; generates urgency-flagged, consequence-aware management recommendations; exposes all outputs via REST API.
- **Harvest Estimation** (IT22266446): Receives disease severity grade and crop growth stage data from this component via API and uses a regression model to estimate expected yield loss as a percentage and LKR financial value, completing the consequence picture for the farmer.

All four modules expose standardized JSON REST APIs consumed by a shared React-based web dashboard accessible to farmers and DoA extension officers.

4.2 Individual Component Architecture

This component operates as a three-layer pipeline within a single containerized FastAPI service:

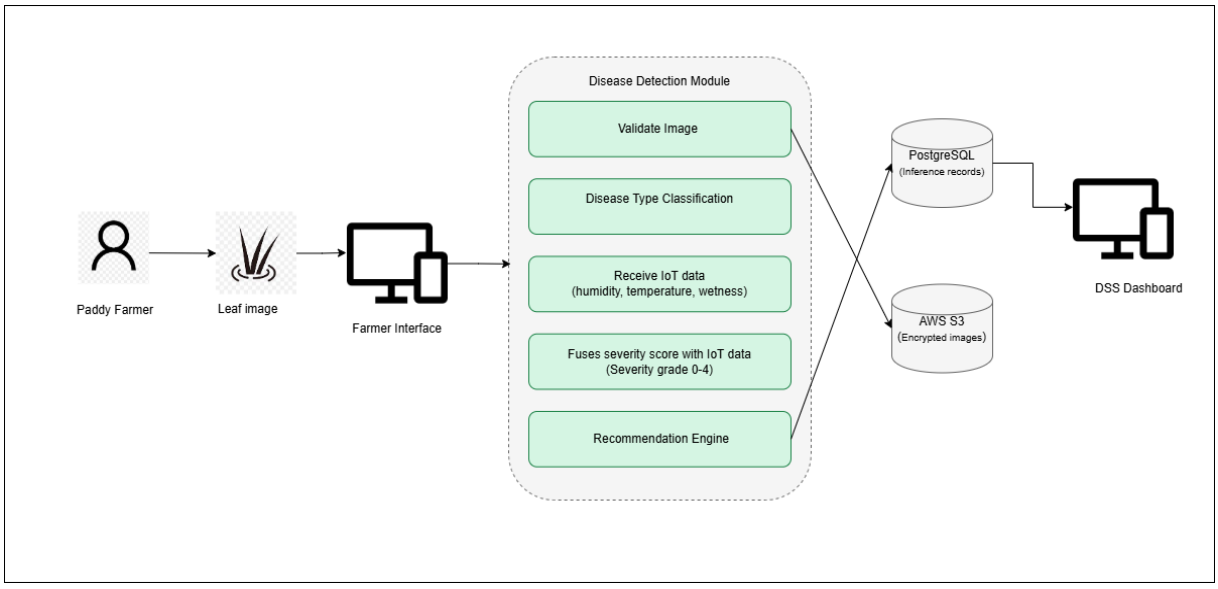


Figure 4: Individual Component Diagram

Data Ingestion and Quality Assurance: Paddy leaf images are accepted via multipart HTTP POST from the mobile or web client, supporting JPEG and PNG formats up to 10 MB. A quality assurance filter at this layer immediately rejects images below 224×224 pixels or with a motion blur detection score above a configured threshold, returning an informative error to the caller before any machine learning inference is invoked. Valid images are passed to Layer 2 alongside the farm identifier and any available farm history retrieved from PostgreSQL.

Pre-processing and Severity Classification: Accepted images are resized to 380×380 pixels EfficientNet-B4's native input resolution and normalised to ImageNet mean and standard deviation. Five-crop Test-Time Augmentation is applied during inference to improve prediction stability under variable field photography conditions. The fine-tuned EfficientNet-B4 model produces a probability vector that is decoded to a disease type and severity grade (0–4), alongside a continuous severity score (0.0–1.0). When model confidence falls below 0.65, the API returns a Low Confidence flag with the top two candidate disease classes, preventing high-stakes decisions from resting on unreliable predictions.

Recommendation Generation and API Response: The detected disease type and severity grade are passed to the rule-based recommendation engine, which consults the DoA-SL guideline rule table to produce a structured recommendation: urgency level, recommended fungicide or cultural practice from the DoA-SL approved input list, and a consequence statement describing the financial risk of delayed action. The complete result includes disease type, severity grade, confidence score, recommendation, urgency flag is assembled into a JSON response and returned via the FastAPI endpoint. The same result is simultaneously written to PostgreSQL with the farm identifier and timestamp for trend analysis and future farm-level history retrieval.

4.3 Security and Scalability Considerations

All API endpoints are protected by JWT authentication. Farm-level inference data is stored in PostgreSQL with row-level security ensuring each farmer can access only their own records. The containerized FastAPI service is deployed on AWS ECS with Docker, configured for horizontal auto-scaling triggered at 70% CPU utilization. Image classification inference is stateless, enabling clean auto-scaling during peak outbreak periods when many farmers submit images simultaneously. All data in transit is encrypted using TLS 1.3 and data at rest uses AES-256. The service is designed to handle a minimum of 100 concurrent image analysis requests while maintaining response times below 8 seconds.

5. USER REQUIREMENTS

5.1 Stakeholder identification

The disease severity estimation component serves four main groups of stakeholders:

- Smallholder paddy farmers: The primary end-users who interact with the system via mobile or web interface. They receive disease alerts and management recommendations in both Sinhala and English, presented in plain, actionable language.
- Department of Agriculture extension officers: Professional users who need exportable diagnostic reports for their advisory records and may use the system to support multiple farmers during field visits.
- Group project integration team: API consumers who require well-documented JSON contracts to integrate this component's output into the shared DSS dashboard and connect it with the IoT, crop recommendation, and growth monitoring modules.
- SLIIT academic supervisor: Evaluates research rigor, novelty, and engineering quality of the component against final-year dissertation standards.

5.2 Functional requirements

Table 3: Functional Requirements

ID	Title	Description
FR01	**Image Capture & Upload**	The system shall accept paddy leaf images via mobile camera capture or file upload from the web interface, accepting JPEG and PNG formats up to 10 MB.
FR02	**Quality Validation**	The system shall reject images below 224×224 resolution or with excessive motion blur and return an informative error message.
FR03	**Disease Classification**	The system shall classify the uploaded image into one of five disease categories: rice blast, brown spot, bacterial leaf blight, sheath blight, or healthy.
FR04	**Severity Grade Output**	The system shall output a severity grade (0–4) and a continuous severity score for the detected disease.
FR05	**Environmental Integration**	The system shall retrieve current humidity, temperature, and leaf wetness readings from IT22178718 and incorporate these into the severity progression calculation.
FR06	**Management Recommendation**	The system shall generate a structured management recommendation including urgency level, recommended intervention, and financial consequence of delayed action.
FR07	**Result History**	The system shall store all inference results, environmental readings, and recommendations in PostgreSQL with timestamps, and allow retrieval for trend analysis.

FR08	**REST API Exposure**	The system shall expose disease class, severity score, progression forecast, harvest loss estimate, and recommendation via FastAPI endpoints for consumption by the DSS dashboard.

5.3 Non-functional requirements

Performance
Performance means the system must give the farmer a result within 5 seconds. This matters because when a farmer uploads a photo (FR01), the system must do several things one after another check the image quality, detect the disease (FR03), grade the severity (FR04), get weather data (FR05), and produce a recommendation (FR06) all within that 5-second limit. If even one step is too slow, the whole thing fails. The API that sends results to the dashboard (FR08) also has to be fast enough to keep the dashboard working in real time. So, performance is not just a nice-to-have, it controls how every part of the system must be built.

Accuracy
Accuracy means the model must correctly identify the disease and its severity at least 85% of the time. This is the most important quality requirement because everything else depends on it. If the model gets the disease wrong in FR03, or gives the wrong severity grade in FR04, then the recommendation in FR06 will also be wrong possibly telling a farmer to just monitor a situation that is critical. The wrong result also gets saved as history in FR07 and sent through the API in FR08, spreading the mistake further. This is why accuracy has to meet the target before the system is launched, not fixed later.

Reliability
Reliability means the system must be available and working 99% of the time during the growing seasons. The API (FR08) is the most affected because the dashboard and other components all depend on it being online. If it goes down during a disease outbreak exactly when farmers need it most nobody can use it. The result storage in FR07 is also affected because if the system keeps going offline, the farm history records become incomplete and trend analysis becomes useless. To meet this requirement, the system uses auto-scaling cloud deployment that handles traffic spikes without crashing.

Usability
Usability means a farmer who has never used a technology app before must be able to upload a photo and get a recommendation in just 3 steps. This affects the upload screen (FR01) and the recommendation output (FR06) the most. The upload must be straightforward a farmer should not need training to find the camera button. The recommendation must be written in plain language not technical terms or numbers that require expert knowledge to understand. The weather data (FR05) also must be collected

automatically so the farmer does not need to type in humidity or temperature values manually. For extension officers, history retrieval (FR07) must also be quick and simple during field visits.

Security

Security means all farm data must be protected with encryption and only accessible to the right people. When a farmer uploads a photo (FR01), that image shows the health of their crop and is sensitive information. It must be encrypted when sent and when stored. The farm history saved in FR07 including every disease result, recommendation, and timestamp is personal data that belongs to each farmer, so the database must be set up so that one farm cannot see another farm's records. The API (FR08) must require a login token so that only the authorized dashboard, IoT module, and harvest estimation component can access the disease detection results.

6. COMMERCIALIZATION PLAN

6.1 Target market and customer persona

The major target market includes the 1.8 million smallholder paddy agriculture households in Sri Lanka, primarily located in the North Central, North-Western, Southern, and Eastern Provinces. The major customer profile includes paddy farmers between the ages of 35 and 60 years, possessing 0.5 to 2 hectares of land, using a simple smartphone device, having a low level of formal agricultural education, and showing high reliance on advice from extension officers.

Secondary target markets include:

(a) Department of Agriculture extension officers requiring digital diagnostic solutions for improving their advisory services

(b) Agro input companies, including fertilizer and fungicide companies, aiming for precise targeting of disease-affected clusters.

(c) Crop insurance companies require objective records of disease severity for adjudicating claims.

6.2 Value proposition

Unlike existing disease detection applications that only confirm whether a disease is present, this system answers the question every farmer genuinely needs answered what should I do right now, and what happens if I wait? The consequence-aware, urgency-graded recommendation output, combined with full localization for Sri Lankan paddy varieties and climatic conditions, provides a differentiated value proposition that no existing product currently offers in the Sri Lankan market.

6.3 Revenue model

- Freemium Mobile Application: Basic disease detection features and fundamental recommendation features are provided for free, while historical trend analytics, 7-day progression forecasting, PDF report generation, and priority user support are available for a monthly fee of 500 LKR for each farm.
- B2B API Licensing: Agro-input businesses and insurance underwriters are charged a query-based fee for using our API, which will enable them to incorporate severity data into their own platforms, with a cost estimate of around 5 LKR for each classification.
- Government Partnership: A proposal for our API to be integrated into the e-Krushi national agricultural advisory portal, provided by DoA-SL, under a government SaaS contract, ensuring a guaranteed source of recurring revenue.
- Dataset Licensing: The annotated Sri Lanka paddy disease dataset will be licensed as a paid product for agricultural research institutions and insurance companies.

6.4 Cost estimation

The estimated monthly running costs will be the AWS EC2 hosting cost (LKR 15,000), the PostgreSQL database cost (LKR 5,000), and the maintenance cost (LKR 20,000), making a total of LKR 40,000. With a 1% premium conversion rate from the addressable population of 500,000 active smallholder farmers in the primary districts, the acquisition of 5,000

premium subscribers will generate a revenue of LKR 2,500,000 on the second month after the launch, thus breaking even. The upside provided by the B2B API adds to the compressing of the break-even period.

6.5 Competitive advantage IP considerations

The main competitive advantage is the locally annotated Sri Lankan paddy disease dataset. The dataset is unique in the world and meets both the DoA-SL and IRRI SES requirements. The technical contributions of the project, which include the CNN-LSTM model and the harvest loss estimation function, will be outlined in a future publication in a journal. The annotated dataset will be maintained as institutional IP by SLIIT and a commercialization license will be granted to the project team.

7. BUDGET AND JUSTIFICATION

The total project budget is divided into different cost categories: cloud computing, cloud hosting for API deployment, database management, field data collection, and hardware for standardized images. All the amounts will be in Sri Lankan Rupees (LKR).

Table 5: Budget Breakdown

Item	Justification	Estimated Cost (LKR)
Cloud GPU compute (AWS EC2 G4dn) model training	DL model training requires GPU compute; estimated 100 hrs across multiple training runs.	12,000
AWS S3 dataset storage (50 GB × 12 months)	Image dataset and model checkpoint storage during research period.	10,000
AWS Lambda inference API hosting (12 months)	Serverless inference endpoint; scales with request volume without fixed server cost.	8,000
Field data collection transport (3 field visits, North Central Province)	Travel to Anuradhapura and Polonnaruwa for farmer-collaborative image collection.	8,000
Leaf image annotation (Roboflow pro plan 6 months)	Professional annotation platform with expert labeling quality control.	12,000
Agricultural expert consultation (2 consultants, 4 sessions)	Severity label verification and recommendation validation by certified officers.	10,000
Printing questionnaires, consent forms, field sheets	Farmer consent forms and usability testing questionnaires.	2,000
Miscellaneous (cables, memory cards, contingency)	Unexpected field or hardware incidentals.	13,000
TOTAL		**75,000 LKR**

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Work Breakdown Structure

The work is organized into five major work packages aligned with the specific objectives:

• WP1 – Dataset Curation and Annotation (Weeks 1–4): Field image collection in three districts; PlantVillage integration; Label Studio pipeline creation; inter-annotator agreement validation; and DVC for version control.

• WP2 – EfficientNet-B4 Model Training (Weeks 4–7): Transfer learning with pre-trained weights on ImageNet; two-stage fine-tuning on Sri Lanka dataset; hyperparameter optimization; and accuracy and F1-score validation on a held-out test set.

• WP3 – CNN-LSTM Progression Model (Weeks 6–9): Temporal sequence dataset creation from historical images and environmental sensor data; CNN-LSTM implementation; and 3-day and 7-day forecast evaluation.

• WP4 – Harvest Loss Engine and API Deployment (Weeks 8–11): Parameterization of loss function from DoA-SL tables; implementation of recommendation rule table; development with FastAPI; containerization with Docker; deployment on AWS ECS; and load test with 50 concurrent users.

• WP5 – Field Pilot and Evaluation (Weeks 11–16): Deployment with 20 farmers and 10 DoA officers in Polonnaruwa district; SUS usability test; accuracy validation with expert-provided ground truth for 50 test plants; bias analysis for each district; and creation of a final report.

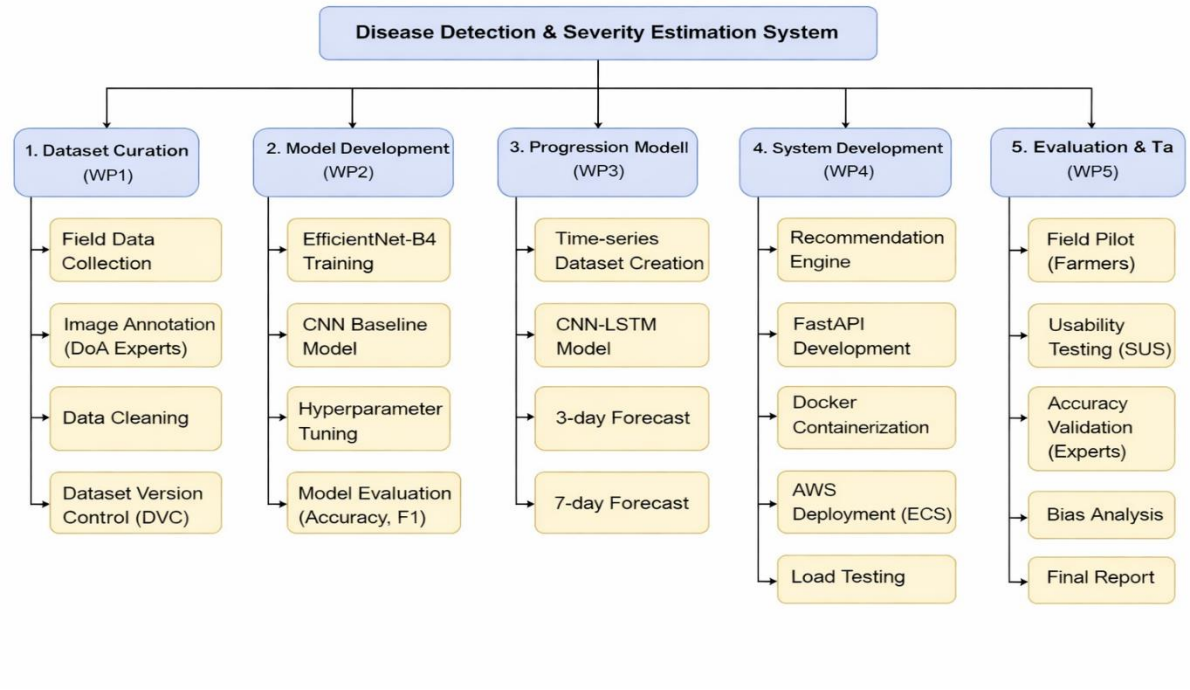


Figure 5: Work Breakdown Structure

8.2 Risk Analysis

Table 5: Risk Analysis

Risk	Probability	Impact	Rating	Mitigation Strategy
Insufficient field images for minority disease-severity classes	Medium	High	**HIGH**	PlantVillage augmentation + Albumentations synthetic generation; re-target 150 images/class minimum as fallback
IoT API unavailable during inference	Low	Medium	**MEDIUM**	Fallback to historical average humidity for farm district; alert flag added to API response
EfficientNet-B4 convergence failure on small dataset	Medium	High	**HIGH**	Layer-wise learning rate decay; 10-epoch frozen backbone pre-training before full fine-tune
DoA annotation expert availability delays	Medium	Medium	**MEDIUM**	Engage 3 independent annotators; majority-vote labelling reduces single-expert bottleneck
Farmer reluctance during field pilot adoption	Low	Medium	**LOW-MED**	Pre-pilot Sinhala-language training session; DoA officer co-facilitation; simplified UI walkthrough

9. GANTT CHART

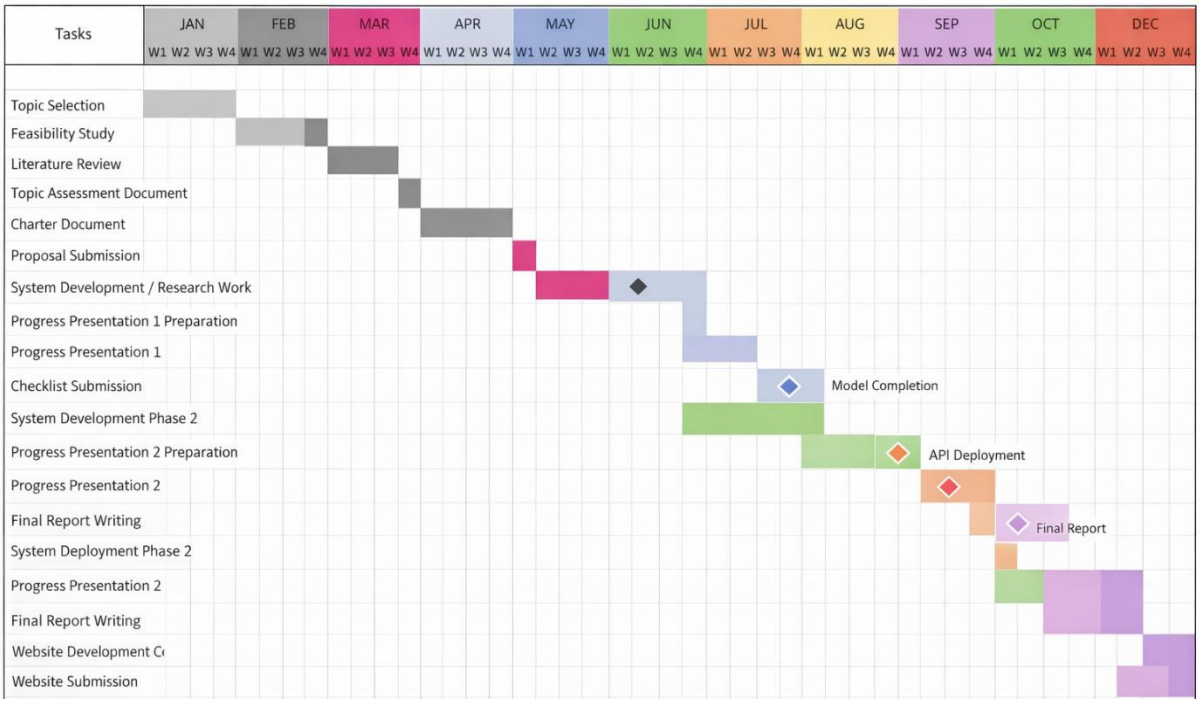


Figure 6: Gantt Chart

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11.APPENDICES

Appendix A: Project TAF Reference

Project ID: R26-IT-066 | Component: IT22261632 | Research Group: SST – Software Systems & Technologies | Specialization: Information Technology | Maha 2025/26 season project initiation.

Appendix B: IRRI Severity Annotation Scale

All disease images are annotated using the IRRI Standard Evaluation System (SES) severity scale: Score 0 = No disease; Score 1–3 (Mild) = ≤10% leaf area affected; Score 4–6 (Moderate) = 11–50% leaf area affected; Score 7–9 (Severe) = >50% leaf area affected. Three independent reviewers annotate each image; Cohen's Kappa ≥0.75 required for label finalisation. DoA-SL extension officers serve as adjudicators for disputed labels.

Appendix C: Ethical Considerations

Field image collection will be conducted with written informed consent from participating farmers. No personally identifiable data will be stored in association with images. Farm location data is stored at district level only (not GPS coordinates) to protect farmer privacy. Data processing complies with the Sri Lanka Computer Crimes Act (2007) and SLIIT ethical research guidelines. SLIIT Faculty of Computing Research Committee ethical clearance will be obtained prior to field data collection and pilot deployment.